

A High-Precision Multi-Factorial Machine Learning Framework for Early Cardiovascular Risk Stratification Using Hemodynamic and Metabolic Composite Markers

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ABSTRACT

Cardiovascular diseases (CVDs) remain the leading cause of global mortality, accounting for 17.9 million annual deaths. Traditional risk calculators (Framingham, SCORE2) often rely on rigid linear scoring heuristics that fail to capture non-linear metabolic interactions and arterial wall stiffness dynamics. In this study, we present CardioGuard, an interpretable end-to-end machine learning framework trained across a diverse cohort of 8,763 patient records with 26 clinical and demographic dimensions. We introduce continuous physiological feature transformations including Pulse Pressure (PP), Mean Arterial Pressure (MAP), Atherogenic Lipid Indices, and a synergistic Metabolic Risk Composite Score. Benchmarked across seven machine learning classifiers, our regularized Logistic Regression model achieved 98.97% test accuracy (ROC-AUC 0.9994, Precision 98.93%, Recall 97.27%), while Gradient Boosting achieved 95.21% accuracy (ROC-AUC 0.9909), and a Soft Voting Ensemble achieved 97.09% accuracy (ROC-AUC 0.9972). The system is deployed as an open-access clinical decision support web application with serverless REST APIs.

Index Terms — Cardiovascular Disease, Heart Attack Prediction, Gradient Boosting, Soft Voting Ensemble, Hemodynamic

1. INTRODUCTION

Coronary artery disease (CAD) and acute myocardial infarction (AMI) are the principal contributors to global cardiovascular morbidity. Over 50% of sudden cardiac death events occur in patients without pre-existing diagnosed symptoms, highlighting an urgent clinical need for early, high-sensitivity screening tools. Conventional risk stratification tools—such as the Framingham Risk Profile and the 2019 ACC/AHA Primary Prevention Guidelines—frequently underestimate risk in younger cohorts and metabolic syndrome phenotypes due to their reliance on static additive points. To overcome these limitations, CardioGuard leverages non-linear machine learning ensembles coupled with domain-guided hemodynamic feature extraction.

2. COHORT CHARACTERIZATION & DATASET ARCHITECTURE

The experimental dataset comprises 8,763 patient records with 26 comprehensive features, capturing physiological vitals, lipid biomarkers, chronic medical history, and behavioral lifestyle metrics. The cohort spans an age range of 22 to 85 years (Mean: 52.4 ± 14.8 years; 65% Male, 35% Female) sampled across 12 countries. The binary target endpoint represents acute heart risk (27.16% positive prevalence, reflecting real-world clinical preventative cardiology screening distributions).

Table 1: Clinical feature taxonomy and domain parameters utilized across the CardioGuard framework.

3. METHODOLOGY & CLINICAL FEATURE ENGINEERING

To elevate classification accuracy from baseline heuristics to diagnostic precision (>95%), we derived continuous physiological markers directly reflecting vascular biomechanics and lipid atherogenicity, implemented in src/preprocessing.py:

- **Pulse Pressure (PP):** $PP = SBP - DBP$
Direct index of large artery stiffness and pulsatile strain (Franklin et al., 1999).
- **Mean Arterial Pressure (MAP):** $MAP = \frac{2 \cdot DBP + SBP}{3}$
Calculates steady organ perfusion pressure (Sesso et al., 2000).
- **Atherogenic Lipid Index:** $Lipid\ Index = \frac{Cholesterol \times Triglycerides}{10,000}$
Quantifies circulating atherogenic lipid accumulation product.
- **Triglyceride-to-Cholesterol Ratio (TCR):** $TCR = \frac{Triglycerides}{Cholesterol + 1}$
Biochemical proxy for small, dense, highly atherogenic LDL particles.
- **Sedentary-to-Active Ratio:** $SAR = \frac{Sedentary\ Hours/Day}{(Exercise\ Hours/Week/7) + 0.1}$
Quantifies metabolic deconditioning and physical inactivity imbalance.
- **Metabolic Risk Composite:**